

Essential numerical range characterization of unitary-similar L -asymptotic limits

A complete answer on separable infinite-dimensional Hilbert space

candidate full separable solution—likely valid

1. Source question and exact scope

Gehér writes after Theorem 4.2 of arXiv:1505.07205 [1]:

“We note that it is an open problem to describe the L asymptotic limits of those operators that are similar to unitary operators and act on infinite dimensional spaces.”

The surrounding theorem and the exact question are visible in the source-page crop below.

namely we will prove the following statement where (i) is only an implications statement, but (ii) is an “if and only if” one. We recall that the minimal element of the spectrum of a self-adjoint operator A is denoted by $\underline{r}(A)$.

THEOREM 4.2.

- (i) *Let $\dim \mathcal{H} = \infty$ and $T \in \mathcal{B}(\mathcal{H})$ be a contraction which is similar to a unitary operator. Then $\dim \ker(A_T - \underline{r}(A_T)I) \in \{0, \infty\}$. Consequently, the condition $\underline{r}(A_T) \in \sigma_e(A_T)$ holds.*
- (ii) *Let $\dim \mathcal{H} = \aleph_0$, $A \in \mathcal{B}(\mathcal{H})$ be a positive, invertible contraction and suppose that the conditions $1 \in \sigma_e(A)$ and $\dim \ker(A - \underline{r}(A)I) \in \{0, \aleph_0\}$ are fulfilled. Then there exists a contraction $T \in \mathcal{B}(\mathcal{H})$ which is similar to a unitary operator and the asymptotic limit of T is exactly A .*

This theorem can be considered as a strengthening of the necessity part in Sz.-Nagy’s similarity theorem. In particular, in the separable case it provides a characterization of asymptotic limits for those contractions which are similar to unitary operators. We note that it is an open problem to describe the L asymptotic limits of those operators that are similar to unitary operators and act on infinite dimensional spaces.

2. Proofs

We solve the problem for separable infinite-dimensional complex Hilbert spaces. This is the standard setting of the source paper and is also the setting in which the diagonal Weyl–von Neumann theorem supplies the decisive construction. No nonseparable extension is claimed.

2. Definitions

A *Banach limit* is a positive norm-one functional $L : \ell^\infty(\mathbb{N}) \rightarrow \mathbb{C}$ which extends the ordinary limit and is invariant under the left shift. If $T \in \mathcal{B}(\mathcal{H})$ is power bounded, its L -asymptotic limit is the positive operator $A_{T,L}$ determined by

$$\langle A_{T,L}x, y \rangle = L_n \langle T^n x, T^n y \rangle. \quad (1)$$

For a unitary U and $B \in \mathcal{B}(\mathcal{H})$, write

$$\langle \Phi_{U,L}(B)x, y \rangle = L_n \langle U^{-n} B U^n x, y \rangle. \quad (2)$$

This is a positive unital linear map, and shift invariance gives $U^* \Phi_{U,L}(B) U = \Phi_{U,L}(B)$.

The essential numerical range is

$$W_e(A) = \{ \lim_k \langle A x_k, x_k \rangle : (x_k) \text{ is an orthonormal sequence} \}.$$

For self-adjoint A it is the interval

$$W_e(A) = [\min \sigma_e(A), \max \sigma_e(A)]. \quad (3)$$

3. Main result

Theorem 1 (complete separable characterization). *Let $\mathcal{H}$ be a separable infinite-dimensional complex Hilbert space, and let $A \in \mathcal{B}(\mathcal{H})$. The following are equivalent.*

1. *For some Banach limit L , there is an operator T similar to a unitary such that $A_{T,L} = A$.*
2. *For every Banach limit L , there is an operator T similar to a unitary such that $A_{T,L} = A$.*
3. *A is positive and invertible, and $1 \in W_e(A)$.*

Moreover, in (2) the same T may be used for all Banach limits. More precisely, whenever (3) holds, there is a bilateral-shift unitary U for which

$$T = A^{-1/2} U A^{1/2} \quad (4)$$

satisfies $A_{T,L} = A$ for every Banach limit L .

Since A is positive and invertible,

$$1 \in W_e(A) \iff 1 \in W_e(A^{-1});$$

indeed, the two essential spectral intervals have endpoints related by inversion. We use the inverse formulation in the construction.

4. Proof intuition

Write $T = SUS^{-1}$. Averaging S^*S along the unitary orbit produces an operator commuting with U . Absorbing its square root into S normalizes that average to the identity. After normalization, A^{-1} is unitarily equivalent to a positive operator B whose unitary-orbit Banach mean is I .

Why must 1 lie in $W_e(B)$? If the essential spectrum sat strictly above or below 1 , a finite-rank correction would make B uniformly larger or smaller than I . Its Banach mean would then force the mean of a positive finite-rank operator to dominate a nonzero multiple of I . That is impossible: testing on N orthonormal vectors gives a lower bound proportional to N , while the trace of the finite-rank operator gives a fixed upper bound.

Conversely, let $B = A^{-1}$ and suppose $1 \in W_e(B)$. Weyl–von Neumann writes $B = D + K$, with D diagonal and K compact. Reorder a carefully chosen infinite part of the diagonal along the positive half of $\mathbb{Z}$ so that all moving averages converge uniformly to 1 ; put all remaining diagonal entries on the negative half. Conjugation by the bilateral shift then has Banach mean I on D . The compact part vanishes under the same averaging because the shifted vectors converge weakly to zero. This gives the construction (4) and, crucially, it works for every Banach limit.

5. Two lemmas

Lemma 2 (finite rank cannot have a uniformly positive orbit mean). *Let U be unitary on an infinite-dimensional Hilbert space, let $F \geq 0$ have finite rank, and let L be a Banach limit. Then $\Phi_{U,L}(F) \not\geq \delta I$ for every $\delta > 0$.*

Proof. If $e_1, \dots, e_N$ are orthonormal, positivity and finite linearity of L give

$$\sum_{j=1}^N \langle \Phi_{U,L}(F) e_j, e_j \rangle = L_n \sum_{j=1}^N \langle F U^n e_j, U^n e_j \rangle \leq \text{Tr } F,$$

because $U^n e_1, \dots, U^n e_N$ is orthonormal for every n . A bound $\Phi_{U,L}(F) \geq \delta I$ would therefore imply $N\delta \leq \text{Tr } F$ for every N , a contradiction. $\square$

Lemma 3 (balanced diagonal arrangement). *Let D be a bounded self-adjoint diagonal operator on a separable infinite-dimensional Hilbert space and let $\lambda \in W_e(D)$. Its diagonal basis can be indexed by $\mathbb{Z}$ so that, writing $De_j = d_j e_j$,*

$$\lim_{N \rightarrow \infty} \sup_{r \geq 0} \left| \frac{1}{N} \sum_{n=0}^{N-1} d_{r+n} - \lambda \right| = 0. \quad (5)$$

Consequently, for the bilateral shift $Ue_j = e_{j+1}$,

$$\Phi_{U,L}(D) = \lambda I \quad (6)$$

for every Banach limit L .

Proof. Put $E = \sigma_e(D)$. First suppose $\lambda \in E$. Choose distinct diagonal entries converging to λ , while reserving infinitely many entries, and place the chosen entries at indices $0, 1, 2, \dots$. Put all remaining entries at the negative indices. Convergence of the positive tail implies (5): after discarding a fixed initial segment every error is small, and the contribution of that segment is $O(N^{-1})$, uniformly in the starting index r .

If $\lambda \notin E$, equation (3) gives points $a, b \in E$ with $a < \lambda < b$. Choose disjoint diagonal subsequences $a_k \rightarrow a$ and $b_k \rightarrow b$, again leaving infinitely many entries unused. Let

$$q = \frac{\lambda - a}{b - a} \in (0, 1), \quad w_n = \lfloor (n+1)q \rfloor - \lfloor nq \rfloor.$$

At positions with $w_n = 0$ place successive a_k ; at positions with $w_n = 1$ place successive b_k . Every interval of length N contains $qN + O(1)$ ones, with an absolute discrepancy at most two. Thus the moving averages of the ideal a/b word converge uniformly to $(1-q)a + qb = \lambda$. The errors $a_k - a$ and $b_k - b$ form a null sequence along the positive positions, so their moving averages converge uniformly to zero by the same fixed-head estimate. This proves (5).

For each fixed j , the sequence d_{j+n} differs only by a finite shift and prefix from the positive tail and is therefore almost convergent to λ . Since $U^{-n}DU^n$ remains diagonal, every Banach limit gives diagonal matrix entries λ and off-diagonal entries zero. Hence (6) holds. $\square$

6. Proof of Theorem 1

Suppose $T = SUS^{-1}$ and $A_{T,L} = A$. Put

$$C = \Phi_{U,L}(S^*S).$$

The two-sided bounds on S^*S pass through the positive unital map $\Phi_{U,L}$, so C is positive and invertible. It commutes with U . Define $R = SC^{-1/2}$. Then

$$T = RUR^{-1}, \quad \Phi_{U,L}(R^*R) = I.$$

Indeed, commutation with U lets the two factors $C^{-1/2}$ pass through $\Phi_{U,L}$, so $\Phi_{U,L}(C^{-1/2}S^*SC^{-1/2}) = C^{-1/2}CC^{-1/2} = I$; the same commutation shows that replacing S by R leaves SUS^{-1} unchanged. Using (1) and (2),

$$A = R^{-*}R^{-1} = (RR^*)^{-1}. \quad (7)$$

In particular A is positive and invertible. Set $B = R^*R$. Polar decomposition shows that B is unitarily equivalent to $RR^* = A^{-1}$, and $\Phi_{U,L}(B) = I$.

We claim $1 \in W_e(B)$. If $\min \sigma_e(B) > 1$, choose c with $1 < c < \min \sigma_e(B)$. Then

$$F = (cI - B)_+$$

has finite rank and $B \geq cI - F$. Applying $\Phi_{U,L}$ yields

$$\Phi_{U,L}(F) \geq (c-1)I,$$

contrary to Lemma 2. If $\max \sigma_e(B) < 1$, choose $\max \sigma_e(B) < c < 1$ and put $F = (B - cI)_+$. Now $B \leq cI + F$, so $\Phi_{U,L}(F) \geq (1-c)I$, the same contradiction. Equation (3) proves the claim. Unitary invariance and inversion now give $1 \in W_e(A)$. This proves (1) $\Rightarrow$ (3).

Conversely assume (3), and set $B = A^{-1}$. Then $1 \in W_e(B)$. By the Weyl-von Neumann theorem for self-adjoint operators on separable Hilbert space [3],

$$B = D + K, \quad (8)$$

where D is diagonal self-adjoint, K is compact self-adjoint, and $\sigma_e(D) = \sigma_e(B)$. Hence $1 \in W_e(D)$. Index the diagonal basis and choose the bilateral shift U as in Lemma 3. It follows that $\Phi_{U,L}(D) = I$ for every Banach limit L .

For the compact part, $U^n x \rightarrow 0$ weakly for every x . Compactness gives $KU^n x \rightarrow 0$ in norm, so

$$\langle U^{-n}KU^n x, y \rangle = \langle KU^n x, U^n y \rangle \rightarrow 0.$$

Therefore $\Phi_{U,L}(K) = 0$ for every L , and (8) gives $\Phi_{U,L}(B) = I$ for every L .

Let $R = B^{1/2} = A^{-1/2}$ and define $T = RUR^{-1}$ as in (4). This T is similar to a unitary and hence power bounded. For every Banach limit,

$$A_{T,L} = R^{-*}\Phi_{U,L}(R^*R)R^{-1} = R^{-1}IR^{-1} = B^{-1} = A.$$

Thus one T works simultaneously for all L , proving (3) $\Rightarrow$ (2). The implication (2) $\Rightarrow$ (1) is immediate.

7. Checks, consequences, and literature boundary

Corollary 4. *If $A > 0$ is invertible and $\sigma_e(A) = \{t\}$, then A is an L -asymptotic limit of an operator similar to a unitary if and only if $t = 1$.*

This agrees with the elementary scalar tests: $A = cI$ can occur only for $c = 1$. Finite-dimensional spaces are deliberately excluded; there the finite-rank trace obstruction disappears, and the source already gives a separate eigenvalue characterization in Theorem 4.2.

Gehér's earlier paper [2] characterizes ordinary strong asymptotic limits of contractions and records the same unitary-similar Banach-limit problem. Kubrusly and Duggal [4] study Banach and Cesàro asymptotic limits and similarity criteria, but the bounded searches conducted for this packet found no essential-numerical-range characterization matching Theorem 1. We therefore claim a candidate solution to the stated problem, not publication-level novelty; a broader priority search remains appropriate.

The sole mathematical scope boundary is separability. The necessity argument is dimension-free for every infinite-dimensional Hilbert space, but the sufficiency proof uses the separable diagonal Weyl–von Neumann theorem and a bilateral sequence indexed by $\mathbb{Z}$. A nonseparable analogue may require additional cardinal or representation-theoretic analysis.

8. Verification summary

The proof was audited at the following stress points: normalization of the similarity, passage of order bounds through $\Phi_{U,L}$, the finite-rank trace contradiction, inversion of the essential spectral interval, uniformity of the balanced moving averages, vanishing of compact conjugates, and the final similarity formula. A numerical script checks the discrepancy and moving-average convergence in a two-point diagonal model. This computation is only a sanity check; Lemmas 2 and 3 contain the proof.

Confidence in mathematical validity within the stated separable scope: **93/100**. Recommended human review: check the balanced diagonal arrangement and the normalization identity first, then conduct a wider operator-algebra novelty search.

References

- [1] Gy. P. Gehér, *Asymptotic behaviour of Hilbert space operators with applications*, arXiv:1505.07205 (2015).

- [2] Gy. P. Gehér, *Asymptotic limits of operators similar to normal operators*, arXiv:1407.0525 (2014).
- [3] J. von Neumann, *Charakterisierung des Spektrums eines Integraloperators*, Actualités Sci. Indust. 229 (1935), 1–20; used in its standard self-adjoint Weyl–von Neumann form.
- [4] C. S. Kubrusly and B. P. Duggal, *Asymptotic limits, Banach limits, and Cesàro means*, arXiv:2010.14740 (2020).